

①

Moukthika

Bio test - 4

Date _____
Page _____

Section - A

1) Widal test

2) Allen's rule

3) Nucleopolyhedrovirus

4) Klüppel's disorder

5) Outcross

6) Oxytocin

7) Zona pellucida

8) Tubectomy

9) AB⁺

10) Trichoderma polysporum

11) Barnacles growing on whales

12) Megasporogenesis

13) Adaptive radiation

14) G-dominance

15) Plasmodium falciparum

②

Section - B

16)

Commensalism

Amensalism

→ Here, one species gets affected positively and the other doesn't get affected at all.
(+ / 0)

→ Here, one species gets affected negatively and the other doesn't get affected at all.
(- / 0)

17)

Statins are blood cholesterol-lowering agents that are commercially extracted from Monscus purpureus.

18)

Vasectomy

Tubectomy

→ It is a male sterilization technique where a small part of the vas deferens is tied up to prevent sperm transport.

→ It is a female sterilization method where a small part of the fallopian tube is tied up to prevent ovum transport.

19) Sickle cell anaemia is caused by a point mutation in the β -globin gene, where adenine is substituted by thymine at the 6th codon of the β -globin mRNA. This results in the replacement of glutamic acid by valine at the 6th position of the β -globin polypeptide chain.

- 20) Low atmospheric pressure at high altitudes lead to low partial pressure of Oxygen, causing hypoxia (body doesn't get enough Oxygen)
- Increasing RBC by secreting erythropoietin
 - Increasing breathing rate & decreasing the binding affinity of hemoglobin for Oxygen.

- 21) (i) Nucellus - $2n$ (Diploid)
 (ii) Microspore tetrad - n (Haploid)
 (iii) Endosperm - $3n$ (Triploid)
 (iv) Synergids - n (Haploid)

22) Incomplete dominance Codominance

- | | |
|---|--|
| • Neither allele is completely dominant | • Both alleles are equally dominant and equally expressed. |
| • Eg: Pink flowers from red & white ones in <u>Mirabilis jalapa</u> . | • Eg: $I^A I^B$ producing AB blood group in humans |

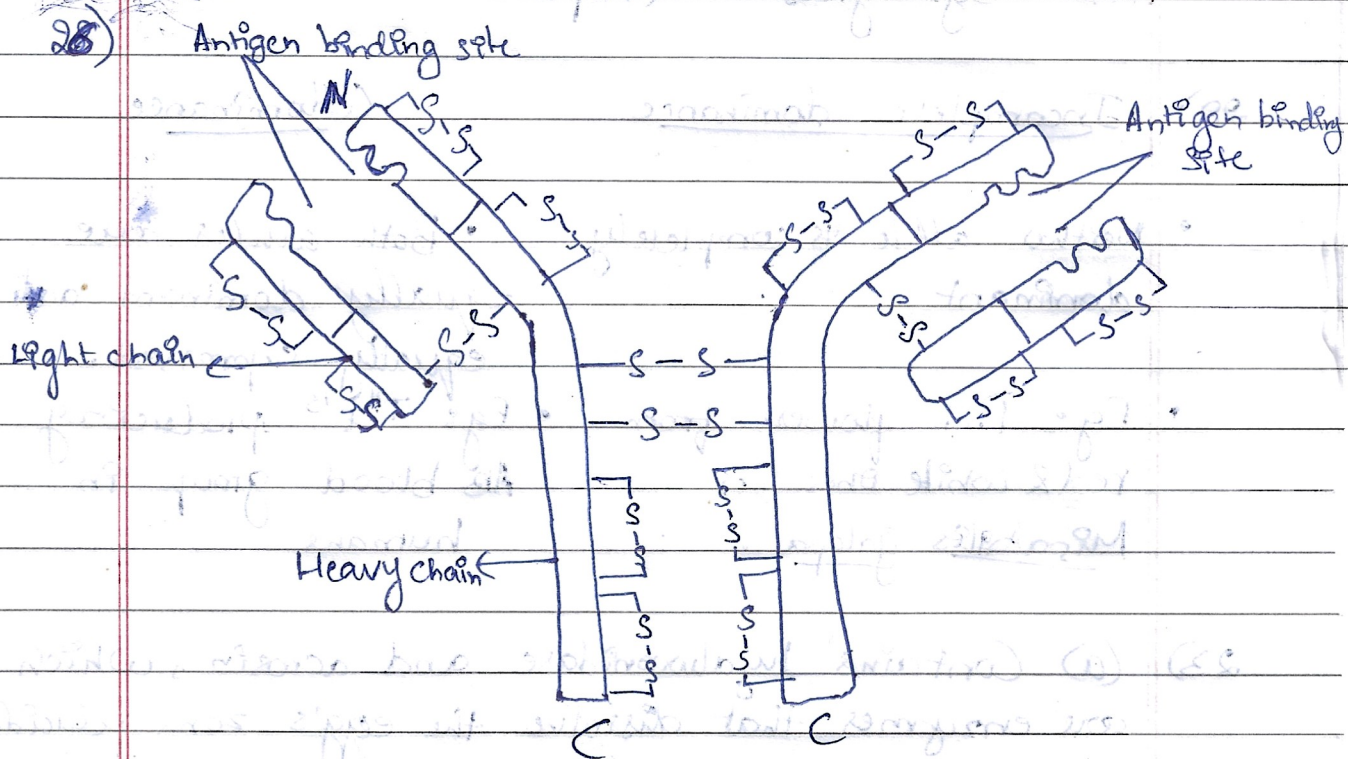
- 23) (a) Contains hyaluronidase and acrosin, which are enzymes that dissolve the egg's zona pelucida to promote fertilisation.
- (b) Synthesise and secrete androgens (testosterone), which in turn facilitate spermatogenesis.

4

24) They are organisms that enrich the nutrient quality of the soil by fixing nitrogen.
Eg of symbiotic nitrogen-fixing bacterium: Rhizobium

25) The development of fruit without prior fertilisation of the ovules, resulting in seedless fruit is called parthenocarpy.
Eg: Banana

Section-C



IgG

5)

27) Reverse transcription: The viral enzyme reverse transcriptase synthesises a single-stranded DNA, followed by a double-stranded viral DNA copy from the RNA.
Integration: Viral DNA enters host nucleus & enters into the host cell's chromosome via the enzyme integrase.
Translation: The integrated proviral DNA directs the production of new viral RNA genomes & viral proteins using host machinery.

28) In birds, females are heterozygous (ZW), determining the gender of the offspring whereas the males are homozygous (ZZ).
In humans on the contrary, males are heterozygous (XY), determining the gender of the baby, while the females are homozygous (XX).

29) Mother: $i i$
Father: $I^A I^B$

$\times$	I^A	I^B
i	$I^A i$	$I^B i$
i	$I^A i$	$I^B i$

$I^A i$ is group A

$I^B i$ is group B

50% chance that child has Blood Group A

50% chance that child has Blood Group B

0% chance that child has group O.

30) Microsporogenesis

- Occurs in the pollen sacs of anthers
- All 4 haploid microspores produced from the PMC are functional.
- Leads to the formation of male gametophytes.

Megasporogenesis

- Occurs in the nucellus of the ovule
- Only 1 megaspore remains functional while the other 3 degenerate.
- Leads to formation of female gametophyte.

31) • Micropylar end (3 cells)

- One central egg cell
- Two synergids cells with cellular thickenings at the micropylar tip called the filiform apparatus.

• Central region (1 cell², 2 nuclei)

- A large central cell containing 2 polar nuclei that fuse to form a 2ndary diploid nucleus.

• Chalazal End (3 cells)

- 3 antipodal cells that play a nutritive role and then degenerate.

32) 1) Inhibition of Ovulation: Prevents follicular development and the LH surge necessary for ovulation

2) Inhibition of Implantation: Makes the endometrium hostile for blastocyst implantation

3) Cervical Mucus alteration: Increasing the thickness of cervical mucus, preventing sperm entry into the uterus.

~~UAA/AN~~

- 33) (a) Chromosomal cause: Trisomy of chromosome 21
2 symptoms: Short stature with a round head, broad & flat face
(b) Chromosomal cause: Monosomy of chromosome X
2 symptoms: Sterile females, webbed neck

- 34) 1) Nutrient absorption: Enhance the absorption of essential minerals from the soil.
2) Disease resistant: Providing resistance to root-borne pathogens
3) Environmental tolerance: Increasing the plant's tolerance to extreme environmental conditions.

35) Homologous

Analogous

- Organs sharing common structural origin but adapting to different functions
Eg: Thorns of Bougainvillea & tendrils of Cucurbita
→ Organs sharing different evolutionary origins but performing similar function
Eg: Sweet potato & potato.

Section - D

36) Option - A

Human host:

- 1) Infection: An infected female Anopheles mosquito bites a human, spreading infectious sporozoites into the bloodstream.
2) Hepatic Schizogony: The sporozoites reach the liver through the blood within half an hour. They multiply

8

asexually within the liver cells, bursting the cells and releasing 1000s of parasites into the blood.

Erythrocytic Schizogony: The released parasites attack RBCs, where they multiply asexually. The infected RBCs rupture, releasing more parasites.

Rupture of RBCs: This is accompanied by the release of a toxic substance called haemozoin. This is responsible for the characteristics like ~~fever~~ recurring high fever.

Gametocytogenesis: After repeated asexual cycles, some parasites differentiate into gametocytes in the RBCs.

Mosquito host:

Transmission to Mosquito: When another Anopheles mosquito drinks the infected human's blood, it ingests the gametocytes along with the blood.

Fertilisation & development: The gametocytes fertilise and mature inside the mosquito's gut. The zygote develops into a motile stage that breaks the gut wall.

Sporogony: These mature into sporozoites which escape from the gut and migrate to the mosquito's salivary glands. ~~later~~

The cycle then repeats.

Question - 37 (Option - A)

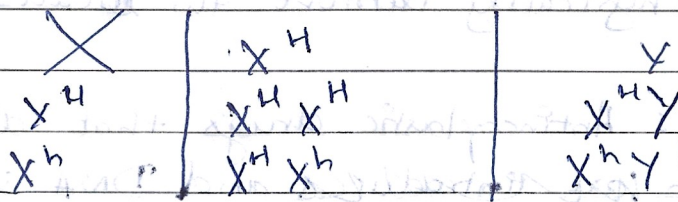
Hemophilia is an X-linked recessive bleeding disorder caused by a deficiency of Hemophilia A or Hemophilia B.

The

9)

Carrier woman: $X^H X^h$

Normal man: $X^H Y$



$X^H X^H$ - Normal daughter $X^H Y$ - Normal son

$X^H X^h$ - Carrier daughter $X^h Y$ - Hemophilic son

→ 50% chance of son being hemophilic

→ 0% chance of daughter being hemophilic
(50% being carrier)

Hemophilia is rare in females as it requires her father to be hemophilic ($X^h Y$) and mother to be either a carrier ($X^H X^h$) or hemophilic ($X^h X^h$).

38) Option - A

Cancer is an uncontrolled proliferation of cells resulting from a loss of normal cell cycle regulation and contact inhibition, capable of spreading throughout the body.

Benign Tumour

Malignant

- Remains confined to its original site
- Slow-growing
- Rarely fatal

- Invades surrounding normal tissues & spreads
- Rapidly-growing
- Life-threatening

Cancer treatment:

- 1) Surgery: Physically remove the localised tumour masses
- 2) Chemotherapy: Antineoplastic drugs that disrupt cell division ~~is~~ introduced and DNA synthesis in rapidly dividing cells is introduced.
- 3) Radiotherapy: High energy ionizing radiation directed to selectively damage and kill tumour cells while leaving the surrounding normal cells undisturbed.

